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MIL-STD-2045-17502-1A
29 July 1994

MILITARY STANDARD

Information Technology
DOD Standardized Profiles AMH2n(D)
Message Handling Systems (MHS)
Military Messaging (P772)

Part 1: MM MHS Service Support



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Foreword

This military standard is approved for use by all Departments and Agencies of the Department of Defense (DOD).

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this MIL-STD should be addressed to the:

Joint Interoperability and Engineering Organization (JIEO)
ATTN: TBBF
Building 283
Fort Monmouth, New Jersey 07703-5613

by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this MIL-STD or by memorandum.

This DOD Standardized Profile (DSP) is a functional standard produced by the Data Communications Protocol Standards (DCPS) Technical Management Panel (DTMP). The MIL-STD-2045 document series was established within the DCPS Standardization Area to allow for the enhancement of commercial standards or the development of standards that are unique to . DTMP functional standards are functional groupings of base standards. Referenced base standards may be commercial, or de facto standards, although International Standards (produced by the International Standards Organization (ISO), the Consultative Committee for International Telephone & Telegraph (CCITT) (now known as International Telecommunication Union (ITU)), and other bodies) are preferred when possible.

The MIL-STD-2045-10000 series, MIL-STD-2045-10000 to MIL-STD-2045-19999 inclusive, will be used to describe how DOD will implement commercial, international, national, federal, or military standards within the functional profile concept to provide required network services. The Government Open Systems Interconnection Profiles (GOSIP) will serve as the base for developing the 10000 series with enhancements, unique military standards, and interim standards being used only when necessary.

The MIL-STD-2045-20000 series, MIL-STD-2045-20000 to MIL-STD-2045-29999 inclusive, will be used to describe DOD enhancements and extensions to existing commercial, international, national, or federal standards.

The MIL-STD-2045-30000 series, MIL-STD-2045-30000 to MIL-STD-2045-39999 inclusive, will be used to describe protocols and services unique to DOD that will not be supported by commercial, international, national, or federal standards.

The MIL-STD-2045-40000 series, MIL-STD-2045-40000 to MIL-STD-2045-49999 inclusive, will be used to document interim standards. Interim standards document protocols and services needed by DOD until these protocols and services are described in either a GOSIP or a MIL-STD-2045-20000 or -30000 series standard.

The MIL-STD-2045-50000 series, MIL-STD-2045-50000 to MIL-STD-2045-59999 inclusive, will be used to describe how DOD will implement commercial, international, national, federal, or military standards within the functional profile concept to provide required network services. The Government Open Systems Interconnection Profiles (GOSIP) will serve as the base for developing the 50000 series with DOD enhancements, unique military standards, and interim standards being used only when necessary. The difference between MIL-STD-2045-10000 series and the MIL-STD-2045-50000 series is that the 50000 series are interim profiles.

Specific details and instructions for establishing a MIL-STD-2045 document, as well as profile development guidelines, are documented in MIL-HDBK-829. DTMP Working Groups shall be responsible for DSP development and informal Service or Agency coordination; the DTMP Plenary shall be responsible for final review and approval.
This document forms part of a multipart DSP for MHS covering Military Messaging requirements. It is outside the scope of the current Taxonomy Framework for International Standardized Profiles. It will correspond to the extensions to that

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taxonomy which will be found in MIL-HDBK-829. This DSP is a content specific profile for the Military Messaging content type referred to as P772 as defined in the Allied Communication Publication (ACP) 123.

This part of MIL-STD 2045-17502 contains two normative annexes and one informative annex.

Annex A (normative)	Elements of Service for ACP 123
Annex B (normative)	Amendments and corrigenda
Annex C (informative)	Concluding Material

For acquisition purposes, where such differences exist, this DSP shall be the controlling document.

The Preparing Activity for this standard is the DTMP. The custodians for the document are identified in the Defense Standardization Program, "Standardization Directory (SD-1)" and are classified in the Federal Supply Classification (FSC) system under Data Communication Protocol Standards (DCPS). Additional information can be obtained from:

Joint Interoperability and Engineering Organization (JIEO)
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Building 283
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Introduction

This DOD Standardized Profile (DSP) is defined within the context of functional standardization, in accordance with the principles specified by ISO/IEC TR 10000, "Framework and Taxonomy of International Standardized Profiles" and MIL-HDBK-829. The context of functional standardization is one part of the overall field of Information Technology (IT) standardization activities - covering base standards, profiles, and registration mechanisms. A profile defines a combination of base standards that collectively perform a specific well-defined IT function. Profiles standardize the use of options and other variations in the base standards to promote system interoperability and to provide a basis for the development of uniform, internationally recognized system tests.

One of the most important roles for a DSP is to serve as the basis for the development of recognized tests. DSPs also guide implementors in developing systems that fit the needs of the US Department of Defense (DOD). DSPs are produced not simply to 'legitimize' a particular choice of base standards and options, but to promote real system interoperability. The development and widespread acceptance of tests based on this and other DSPs is crucial to the successful realization of this goal.

This part of MIL-STD 2045-17502 provides a functional description and specification of MM-specific MHS service support and associated functionality as covered by the AMH2n(D) set of profiles. It identifies what additional service support and functionality can be supported by each type of MHS component in an MM environment (i.e., covering the services supported by an MM UA, plus any additional MTA and MS aspects), divided into basic requirements and zero or more optional functional groups. Such specifications are in many cases applicable to more than one MHS protocol or are otherwise concerned with component functionality which although it can be verified via protocol, is not just related to protocol support. The specification of this part is therefore designed for reference by the following parts (which specify conformance requirements by protocol for each MHS component) and is additional to the protocol-specific requirements specified in those parts. Thus, although this part contains normative requirements, there is no separate conformance to this part (i.e., it is not identified in the MHS taxonomy) since such requirements are only significant when referenced in the context of a particular protocol profile.

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Information technology - DOD Standardized Profiles AMH2n(D) - Message Handling Systems -

Part 1 : Military Messaging Service Support

1 Scope

1.1 General

This part of MIL-STD 2045-17502 contains the overall specifications of the support of MHS Elements of Service and other aspects of MHS functionality which are specific to the Military Messaging (P772) environment which are generally not appropriate for consideration only from the perspective of a single MHS protocol. These specifications form part of the Military Messaging application functions, as defined in the parts of MIL-STD 2045-17502, and are based on the Common Messaging content type-independent specifications in MIL-STD 2045-17501. Such specifications are in many cases applicable to more than one MHS protocol or are otherwise concerned with component functionality which, although it can be verified via protocol, is not just related to protocol support. They are therefore designed to be referenced in the MHS Military Messaging application profiles MIL-STD 2045-17502-2 (AMH21(D)), MIL-STD 2045-17502-3 (AMH22(D)), MIL-STD 2045-17502-4 (AMH23(D)) and MIL-STD 2045-17502-5 (AMH24(D)), which specify the support of specific MHS protocols and associated functionality.

This DSP makes use of the Common Messaging MIL-STD 2045-17501. It specifies the additional requirements needed to support the content type for Military Messaging (P772) as used in ACP 123.

The specifications in this part of MIL-STD 2045-17502 are divided into **basic requirements**, which are required to be supported by all MM MHS implementations claiming conformance to this profile, and a number of optional **functional groups**, which cover significant discrete areas of related functionality which are not required to be supported by all implementations.

1.2 Position within the taxonomy

This part of MIL-STD 2045-17502 is the first part, as common text, of a multipart DSP for AMH2(D) Military Message Handling Systems. The multipart DSP consists of the following parts:

Part 1 - Military MHS service support

Part 2 - AMH21(D) - MM Content (P772)

Part 3 - AMH22(D) - MM Requirements for Message Transfer (P1)

Part 4 - AMH23(D) - MM Requirements for MTS Access (P3)

Part 5 - AMH24(D) - MM Requirements for MS Access (P7)

This DSP shall be combined with the multipart DSP identified as "AMH1n(D), Message Handling Systems - Common Messaging" (see also ISO/IEC TR 10000-1, 8.2 for the definition of multipart ISPs).

The multipart AMH1(D) DSP consists of the following parts:

Part 1 - MHS Service Support

Part 2 - Specification of Remote Operations Service Element (ROSE), Reliable Transfer Service Element (RTSE), and Association Control Service Element (ACSE), Presentation and Session Protocols for use by MHS

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Part 3 - AMH11(D) - Message Transfer (P1)

Part 4 - AMH12(D) - MTS Access (P3)

Part 5 - AMH13(D) - MS Access (P7)

It may be combined with any approved T-Profiles (see ISO/IEC TR 10000) specifying the OSI connection-mode Transport service.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of this part of MIL-STD 2045-17502. At the time of publication, the editions indicated were valid. All documents are subject to revision, and parties to agreements based on this part of MIL-STD 2045-17502 are warned against automatically applying any more recent editions of the documents listed below, since the nature of references made by DSPs to such documents is that they may be specific to a particular edition. Members of IEC and ISO maintain registers of currently valid International Standards and ISPs, and CCITT maintains published editions of its current Recommendations.

Amendments and corrigenda to the base standards referenced are listed in annex B.

NOTE - References in the body of this part of MIL-STD 2045-17502 to specific clauses of ISO/IEC documents shall be considered to refer also to the corresponding clauses of the equivalent CCITT Recommendations (as noted below) unless otherwise stated.

2.1 Government Documents

2.1.1 Specifications, Standards, and Handbooks

MIL-STD 2045-17501-1: *November 1994, Information technology - Standardized Profiles - Message Handling Systems - Common Messaging - Part 1: MHS Service Support.*

MIL-HDBK 829A, *Guidelines for Data Communications Protocol Standards (DCPS) Standardized Profiles (DSPs) July 1994.*

DOD activities may obtain copies of directives through their own publication channels or from the Single Stock Point, Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Other federal agencies and the public may purchase copies from the U.S. Department of Commerce, National Technical Information Service, 5285 Port Royal Road, Springfield, VA 22161.

2.1.2 Other Government Documents, drawings, and publications

ACP 123: *Common Messaging Strategy and Procedures, June 1994.*

2.2 Non-Government Publications

ISO 8613-1: 1993, *Information processing - Text and office systems - Open Document Architecture (ODA) and interchange format Part 1: Introduction and general principles.*

ISO/IEC TR 10000-1: 1992, *Information technology - Framework and taxonomy of International Standardized Profiles - Part 1: Framework.*

ISO/IEC TR 10000-2: 1992, *Information technology - Framework and taxonomy of International Standardized Profiles - Part 2: Taxonomy.*

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ISO/IEC 10021-1: 1990, *Information technology - Text Communication - Message-Oriented Text Interchange Systems (MOTIS) - Part 1: Service Overview. [see also CCITT Recommendation X.400(1992)]*

ISO/IEC 10021-2: 1990, *Information technology - Text Communication - Message-Oriented Text Interchange Systems (MOTIS) - Part 2: Overall Architecture. [see also CCITT Recommendation X.402(1992)]*

ISO/IEC 10021-4: 1990, *Information technology - Text Communication - Message-Oriented Text Interchange Systems (MOTIS) - Part 4: Message Transfer System: Abstract Service Definition and Procedures. [see also CCITT Recommendation X.411(1992)]*

ISO/IEC 10021-5: 1990, *Information technology - Text Communication - Message-Oriented Text Interchange Systems (MOTIS) - Part 5: Message Store: Abstract Service Definition. [see also CCITT Recommendation X.413(1992)]*

ISO/IEC 10021-7: 1990, *Information technology - Text Communication - Message-Oriented Text Interchange Systems (MOTIS) - Part 7: Interpersonal messaging system. [see also CCITT Recommendation X.420(1992)]*

ISO/IEC 10610-1: 1993, *Information technology - International Standardized Profile FOD11 - Open Document Format: Simple document structure - Character content architecture only - Part 1: Document Application Profile (DAP).*

ISO/IEC 10611-1: 1994, *Information technology - International Standardized Profiles - Message Handling Systems - Common Messaging - Part 1: MHS Service Support.*

CCITT Recommendation X.400(1992), *Message handling system and service overview.*

CCITT Recommendation X.402(1992), *Message handling systems: Overall architecture.*

CCITT Recommendation X.411(1992), *Message handling systems: Message transfer system: Abstract service definition and procedures.*

CCITT Recommendation X.413(1992), *Message handling systems: Message store: Abstract service definition.*

CCITT Recommendation X.420(1992), *Message handling systems: Interpersonal messaging system.*

(Application for copies of these documents should be addressed to the American National Standards Institute, 11 West 42nd Street, NY, NY 10036 or to ISO, Van Demonstrate 94, 1013 CN Amsterdam, Netherlands.)

2.3 Order of precedence

In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 Definitions

For the purposes of this part of MIL-STD 2045-17502, the following definitions apply.
Terms used in this part of MIL-STD 2045-17502 are defined in the referenced base standards. In addition, the following terms are defined.

3.1 General

MM base standards : the base standards referred to in this Military Messaging profile consist of both the X.400 base standards and ACP 123 which provides the definition of the MM content type.

Basic requirement : an Element of Service, protocol element, procedural element or other identifiable feature specified in the MM base standards which is required to be supported by all MM MHS implementations claiming support to this profile.

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Functional group : a specification of one or more related Elements of Service, protocol elements, procedural elements or other identifiable features specified in the base standards which together support a significant optional area of MHS functionality.

NOTE - A functional group can cover any combination of MHS features specified in the base standards for which the effect of implementation can be determined at an external interface -i.e., via a communications protocol (other forms of exposed interface, such as a standardized API, are outside the scope of this version of MIL-STD 2045-17502).

3.2 Support classification

To specify the support level of Elements of Service for this part of MIL-STD 2045-17502, the following terminology is defined.

mandatory support (m) :

for origination:

For MT and MS Elements of Service:

a service provider (i.e. an MTA or MS) shall be able to make the Element of Service available to a service user in the role of originator; a service user (i.e. a UA) shall be able to use the Element of Service in the role of originator.

For MM Elements of Service:

a service provider (i.e. an MM UA) shall implement all the procedures specified in the base standards which are associated with the provision of the Element of Service, including use of any corresponding MT or MS Element(s) of Service, as appropriate; where specified in the base standards, a service provider shall make the Element of Service available to the service user in the role of originator; in all cases, it shall be stated in the DSPICS whether the Element of Service is made available to the service user and, if so, how this is achieved.

for reception:

For MT and MS Elements of Service:

a service provider (i.e. an MTA or MS) shall be able to make the Element of Service available to a service user in the role of recipient; a service user (i.e. a UA) shall be able to use the Element of Service in the role of recipient.

For MM Elements of Service:

a service provider (i.e. an MM UA) shall implement all the procedures specified in the base standards which are associated with the provision of the Element of Service, including use of the corresponding MT or MS Element of Service, as appropriate; where specified in the base standards, a service provider shall make the Element of Service available to the service user in the role of recipient; in all cases, it shall be stated in the DSPICS whether the Element of Service is made available to the service user and, if so, how this is achieved.

optional support (o) : an implementation is not required to support the Element of Service. If support is claimed, then the Element of Service shall be treated as if it were specified as mandatory support.

conditional support (c) : the Element of Service shall be supported under the conditions specified in this part of MIL-STD 2045-17502. If these conditions are met, the Element of Service shall be treated as if it were specified as mandatory support. If these conditions are not met, the Element of Service shall be treated as if it were specified as optional support (unless otherwise stated).

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out of scope (i) : the Element of Service is outside the scope of this part of MIL-STD 2045-17502 -i.e., it will not be the subject of a DSP conformance test. However, the handling of associated protocol elements may be specified separately in the subsequent parts of this DSP.

not applicable (-) : the Element of Service is not applicable in the particular context in which this classification is used.

prohibited (x) : the Element of Service may not be used in an implementation claiming conformance to this profile.

4 Abbreviations and Acronyms

ACP	Allied Communication Publication
ACP127	ACP 127 Interworking
AMH	Application Message Handling
ASN.1	Abstract Syntax Notation One
ACSE	Association Control Service Element
CCITT	Consultative Committee for International Telephone & Telegraph
CV	Conversion
DCPS	Data Communications Protocol Standards
DDA	Domain Defined Attributes
DISA	Defense Information Systems Agency
DMS	Defense Messaging System
DOD	Department of Defense
DTMP	DCPS Technical Management Panel
DSP	DOD Standardized Profile
DSPICS	DOD Standardized Profile Implementation Conformance Statement
EoS	Element of Service
FG	Functional group
IEC	International Electrotechnical Commission
ISO	International Standards Organization
ISP	International Standardized Profile
ITU	International Telecommunication Union
ITU-T	ITU Telecommunication Standardization Sector
JIEO	Joint Interoperability and Engineering Organization
MHS	Message Handling Systems
MM	Military Messaging (Message)
MOTIS	Message-Oriented Text Interchange Systems
MS	Message Store
MSP	Message Security Protocol
MT	Message Transfer
MTA	Message Transfer Agent
MTS	Message Transfer System
OSI	Open Systems Interconnection
PD	Physical Delivery
PDAU	Physical Delivery Access Unit
ROSE	Remote Operations Service Element
RTSE	Reliable Transfer Service Element
SEC	Security
UA	User Agent

Support level for Elements of Service (see 3.2):

m	mandatory support
o	optional support
c	conditional support
i	out of scope
-	not applicable
x	prohibited

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5 Conformance

No conformance requirements are specified in this part of MIL-STD 2045-17502.

NOTE - This part of MIL-STD 2045-17502 is a reference specification of the basic requirements and functional groups covered by the AMH2n(D) set of profiles and is additional to the protocol-specific requirements specified in the following parts of MIL-STD 2045-17502. Although this part of MIL-STD 2045-17502 contains normative requirements, there is no separate conformance to this part (i.e., it is not identified in the MHS taxonomy) since such requirements are only significant when referenced in the context of a particular protocol.

Conformance requirements are specified by protocol for each MHS component in the following parts of MIL-STD 2045-17502 with reference to the specifications in this part. Support of the functionality as specified in this part may only be verifiable where the effect of implementation can be determined at a standardized external interface - i.e. via a standard OSI communications protocol. Further, the provision of Elements of Service and other functionality at a service interface will not necessarily be verifiable unless such interface is realized in the form of a standard OSI communications protocol. Other forms of exposed interface (such as a human user interface or a standardized API) may be provided, but are not required for conformance to this version of MIL-STD 2045-17502.

6 Basic requirements

Annex A specifies the basic requirements for support of MHS Elements of Service (EoS) for conformance to MIL-STD 2045-17502 - i.e., the level of support required by all MM MHS implementations, as appropriate to each type of MHS component - i.e. MTA, MS or UA (as MTS-user or MS-user, as relevant).

It shall be stated in the DSPICS which encoded information type values are supported.

7 Functional groups

Annex A also specifies any additional requirements for support of MHS EoS if support of an optional functional group (FG) is claimed, as appropriate to each type of MHS component (i.e., MTA or MTS-user). The following clauses summarize the functionality supported by each of the optional FGs and identify any particular requirements or implementation considerations which are outside the scope of formal conformance to MIL-STD 2045-17502. A summary of the functional groups, identifying which may be supported (Y) and which are not applicable (N) for each type of MHS component (i.e. MTA, MS, UA - whether as MTS-user or as MS-user is not distinguished), is given in the following table.

The FGs defined here are functional groups with regard to this content dependent MM profile. Any of the FGs identified in the Common Messaging MIL-STD 2045-17501 which are not optional FGs in this profile are not identified as FGs here.

The following clauses summarize the functionality supported by each of the optional FGs and specifies which FGs must be supported at the MTS level to support this DSP.

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Table 1 - Summary of AMH2n(D) optional functional groups

Functional Group	MTA	MS	UA
MM Conversion (CV)	Y	N	Y
MM Physical Delivery (PD)	Y	N	Y
MM Security (SEC)	Y	Y	Y
ACP 127 Interworking (ACP127)	N	Y	Y

7.1 MM Conversion (CV)

The MM Conversion FG covers support of those EoS which provide the functionality required to perform the action of message body part conversion. Support of the MM CV FG is applicable to an MTA or a UA. Support of the MM CV FG by either an MTA or UA covers support of the Explicit Conversion EoS only.

NOTE - Support of EoS associated with conversion prohibition is a basic MTA requirement, but this does not imply a capability to perform conversion.

A UA implementation conforming to the MM CV FG shall support use of the MT Explicit Conversion EoS and the MT Conversion Prohibition in Case of Loss of Information EoS. It shall be stated in the PICS which encoded information type conversions the UA can request.

An MTA implementation conforming to the MM CV FG shall conform to the Common Messaging CV FG as specified in MIL-STD 2045-17501 in an MM context (i.e. the ability to perform conversion of MM content is required).

7.2 MM Physical Delivery (PD)

The MM Physical Delivery FG is concerned with access to physical delivery (i.e. postal, courier, etc.) services. The MM PD FG comprises two separate and distinct parts:

support of PD EoS on origination and submission;

support of a co-located physical delivery access unit (PDAU).

Support of PD EoS on submission is applicable for either an MTA or a UA. Support of a PDAU is only applicable for an MTA.

An implementation conforming to the MM PD FG shall conform to the Common Messaging PD FG as specified in MIL-STD 2045-17501. An MM UA conforming to the MM PD FG shall also make the PD EoS available to the MHS user for origination. There are no additional requirements for an MTA in an MM environment.

7.3 MM Security (SEC)

Message Security Protocol (MSP) will provide the basic security for Military Messaging in the US. Therefore support for any of the classes in the Security FG as defined in the Common Messaging SEC FG as specified in MIL-STD 2045-17501 is optional in this MM DSP.

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7.4 ACP 127 Interworking (ACP127)

The ACP 127 Interworking FG covers interworking between implementations conforming to MMHS and the older messaging system following ACP 127 guidelines. Interworking takes place by way of a gateway doing conversion between the two formats. However, if this FG is supported, the MMHS implementation supports those optional services and protocol elements in P772 whose sole purpose is to provide interworking for ACP 127 messages during the transition.

This FG requires support on reception for these services including User Interface display requirements. These services even in the case of the FG will not be originated by the MM-UA but may instead be added at the gateway during conversion and information returned with the gateway notification response.

It is recommended that support for this FG be a configurable option, so it may be turned off when no longer required.

8 Naming and addressing

Implementations shall support naming and addressing capabilities as specified in clause 8 of the Common Messaging MIL-STD 2045-17501-1. In addition, implementations will support the Domain Defined Attributes (DDA) acp-plad and acp-ri for both origination and reception. (Support of these DDAs to identify the UA itself is not required.)

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Annex A

(normative)

ELEMENTS OF SERVICE FOR ACP 123

In the event of a discrepancy becoming apparent in the body of this part of 2045-17502 and the tables in this annex, this annex is to take precedence.

This annex specifies the support constraints and characteristics of what shall or may appear in the implementation columns of a DSPICS of ACP 123. This annex is broadly based on the current version of ISO/IEC ISP for AMH2 (IPMS).

In each table, the "Profile" column reflects the level of support required for conformance to MIL-STD 2045-17502 - i.e. the minimum level of support required by all MM UA implementations supporting this profile (see clause 6). The "Functional Group" column specifies any additional support requirements if support of an optional functional group is claimed (see clause 7). Each column is then further subdivided into support for origination ("Orig") and reception ("Rec".) as defined in 3.2, together with the abbreviated name of the functional group ("FG") in the case of the second column.

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A.1 MT Elements of Service

The requirements for support of MT EoS by an MTA are as specified in clause A.1 of MIL-STD 2045-17501-1. The following tables specify the requirements for use of such services by an MTS-user in an MM environment (i.e. MM UA) for conformance to MIL-STD 2045-17502.

Table A.1 - Elements of Service Belonging to The Basic MM Service (MT EoS)

Element of Service	Profile		Functional Group		
	Orig.	Rec.	FG	Orig.	Rec.
Access Management[1]	m	m			
Content Type Indication[2]	m	m			
Converted Indication	–	m			
Delivery Time Stamp Indication	–	m			
Message Identification	m	m			
Non-delivery Notification	m	–			
Original Encoded Information Types Indication	m	m			
Submission Time Stamp Indication	m	m			
User/UA Capabilities Registration[1]	–	m			

NOTES

1 Implementation of this EoS is a local matter in the case of a co-located MTS-user and will need to be performed using trusted functionality when implemented in combination with security.

2 At a minimum, an implementation must support the content type for MM (P772) which is identified by id-mmhs-content OBJECT IDENTIFIER ::={iso(1)identified-orgnaization(3)nato(26)nato-standard(0)MMHS(4406)object-ids(0)content-type(4)mm88(1)}

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Table A.2 - MM Optional User Facilities (MT EoS)

Element of Service	Profile		Functional Group		
	Orig.	Rec.	FG	Orig.	Rec.
Additional Physical Rendition	o	–			
Alternate Recipient Allowed	m	–			
Alternate Recipient Assignment[2]	–	–			
Basic Physical Rendition	o	–	PD	m	
Content Confidentiality[1]	o	o			
Content Integrity[1]	o	o			
Conversion Prohibition	m	m			
Conversion Prohibition in Case of Loss of Information	o	o	CV	m	
Counter Collection	o	–	PD	m	
Counter Collection with Advice	o	–			
Deferred Delivery[11]	o	–			
Deferred Delivery Cancellation[11]	c[7]	–			
Delivery Notification	m	–			
Delivery via Bureaufax Service	o	–			
Designation of Recipient by Directory Name[9]	m	-			
Disclosure of Other Recipients	o	m			
DL Expansion History Indication	–	m			
DL Expansion Prohibited	m[3]	–			
EMS (Express Mail Service)	o	–	PD	m	
Explicit Conversion	o	–	CV	m	
Grade of Delivery Selection	m	m			
Hold for Delivery	–	c[4]			
Implicit Conversion	–	–			
Latest Delivery Designation	m	–			
Message Flow Confidentiality	i	i			
Message Origin Authentication[1]	o	o			

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Table A.2 - MM Optional User Facilities (MT EoS)

Element of Service	Profile		Functional Group		
	Orig.	Rec.	FG	Orig.	Rec.
Message Security Labelling[1]	o	o			
Message Sequence Integrity[9]	o	o			
Multi-destination Delivery	m	–			
Non-repudiation of Delivery[1]	o	o			
Non-repudiation of Origin[1]	o	o			
Non-repudiation of Submission[1]	i	–			
Ordinary Mail	o	–	PD	m	
Originator Requested Alternate Recipient	m	–			
Physical Delivery Notification by MHS	o	–			
Physical Delivery Notification by PDS	o	–			
Physical Forwarding Allowed	o	–	PD	m	
Physical Forwarding Prohibited	m	–	PD	m	
Prevention of Non-delivery Notification	m[10]	–			
Probe[5]	ix	–			
Probe Origin Authentication[1]	i	–			
Proof of Delivery[1]	o	o			
Proof of Submission[1]	i	–			
Redirection Disallowed by Originator	m[3]	–			
Redirection of Incoming Messages	–	m			
Registered Mail	o	–			
Registered Mail to Addressee in Person	o	–			
Report Origin Authentication[1]	i	i			
Request for Forwarding Address	o	–			
Requested Preferred Delivery Method	o	–			
Restricted Delivery	–	i			
Return of Content	ix[12]	–			

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Table A.2 - MM Optional User Facilities (MT EoS)

Element of Service	Profile		Functional Group		
	Orig.	Rec.	FG	Orig.	Rec.
Secure Access Management	m	m			
Special Delivery	o	_	PD	m	
Undeliverable Mail with Return of Physical Message	o	_	PD	m	
Use of Distribution List	m[6]	_			

Notes

1 Support is according to the security class for which support is claimed - see clause A.1 of AMH1n(D) Part 1. MSP will provide the basic security for Military Messaging in the US. Therefore support for any of the classes in the Security FG is optional in this MM DSP, therefore the SEC FG requirements are not enumerated in this table.

2 The method by which an alternate recipient is specified to the MTA is outside the scope of this DSP

3 Support of this EoS has been made mandatory as the default is allowed (the capability to generate both the "prohibited" and the "allowed" value is required).

4 Support of the Hold for Delivery EoS is mandatory when using the P3 protocol. Implementation is a local matter in the case of a co-located MTS-user and may be dependent on local policy.

5 Although support for Probes is required for MTAs in the base X.400 standard, it is recommended that support by MTS-users is not required. This profile further makes Probes dynamically prohibited.

6 Use of Distribution List on submission is always possible as DLs cannot be distinguished from other O/R addresses

7 If Deferred Delivery is supported, Deferred Delivery Cancellation must also be supported.

8 Origination of the Designation of Recipient by Directory Name EoS implies (for those recipients who have been assigned Directory Names in a Directory Service supported by the originating management domain) that the Directory Name may be specified for recipients. Reception of this service implies display (to the user) of any Directory Names contained in the message.

9 Not part of SEC FG because it requires bilateral agreement to work.

10 The ability to request this service on a particular message may be dependent on the precedence requested.

11 Messages should be held in the originating MTA to provide support for Deferred Delivery and Deferred Delivery Cancellation.

12 Prohibition of this EoS means that bit 3 of the *per-message-indicators* envelope field shall be cleared (zero) and that bit 2 of the *notification-requests* heading field shall be cleared. It implies that the *returned-mm* notification field shall never be originated.

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A.2 MS Elements of Service

A Message Store is optional in the MM profile, however, if one is implemented the following Elements of Service apply.

The requirements for support of MS EoS by an MS are as specified in clause A.2 of MIL-STD 2045-17501-1. The following tables specify the requirements for use of such services by an MS-user in an MM environment (i.e. MM UA) for conformance to MIL-STD 2045-17502.

In the following tables, the "Profile" column reflects the basic requirements for conformance to MIL-STD 2045-17502 - i.e. the minimum level of support required by all MM UA implementations conforming to this profile (see clause 6). The "Functional Group" column specifies any additional support requirements if support of an optional functional group is claimed (see clause 7), together with the abbreviated name of the functional group ("FG").

Table A.3 - Base Message Store

Element of Service	Profile	Functional Group	
	UA	FG	UA
MS Register	m		
Stored Message Deletion	m		
Stored Message Fetching	m		
Stored Message Listing	m		
Stored Message Summary	m		

Table A.4 - MS Optional User Facilities

Element of Service	Profile	Functional Group	
	UA	FG	UA
Stored Message Alert	m		
Stored Message Auto-forward	m		

A.3 MM-specific Elements of Service

Note: Although MM was developed using IPMS as a base and adding extensions where necessary to support Military requirements, the base standard for the definition of the MM content type is ACP 123 (harmonized with STANAG 4406). Therefore some of the base standard classifications for an EoS may not always agree with the classification of the same EoS used in an IPMS context.

The following tables specify the requirements for support of MM-specific Elements of Service by an MTS-user in an MM environment (i.e. MM UA) for conformance to MIL-STD 2045-17502.

In the following tables, the "Profile" column reflects the basic requirements for conformance to MIL-STD 2045-17502 - i.e. the minimum level of support required by all MM UA implementations claiming conformance to this profile (see clause 6). The "Functional Group" column specifies any additional support requirements if support of an optional functional group is claimed (see clause 7). Each column is then further subdivided into support for

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origination ("Orig") and reception ("Rec") as defined in 3.2, together with the abbreviated name of the functional group ("FG") in the case of the second column.

Table A.5 - Elements of Service Belonging to the Basic MM Service (MM EoS)

Element of Service	Profile		Functional Group		
	Orig.	Rec.	FG	Orig.	Rec.
Copy Precedence	m	m			
MM-message-identification	m	m			
Primary Precedence	m	m			
Subject Indication	m	m			
Typed Body	m	m			

Table A.6 - MM Optional User Facilities (MM EoS)

Element of Service	Profile		Functional Group		
	Orig.	Rec.	FG	Orig.	Rec.
ACP127 Message Identifier	o	m	ACP127	m	
ACP127 Notification Request	o	–	ACP127	m	
ACP127 Notification Response	o	m	ACP127	m	
Authorizing Users Indication	m	m			
Auto-forwarded Indication	o	m			
Blind Copy Recipient Indication	m	m			
Body Part Encryption Indication	o	m			
Clear Service	o	m			
Codress Message Indicator	o	m	ACP127	m	
Corrections	o	m	ACP127	m	
Cross-referencing Indication	m	m			
Distribution Code	m	m			
Exempted Addresses	m	m			
Expiry Date Indication	m	m			
Extended Authorization Info	m	m			
Forwarded Message Indication	m	m			

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Table A.6 - MM Optional User Facilities (MM EoS)

Element of Service	Profile		Functional Group		
	Orig.	Rec.	FG	Orig.	Rec.
Handling Instructions	o	m	ACP127	m	
Importance Indication[3]	i	i			
Incomplete Copy Indication	m	m			
Language Indication	m	m			
Message Instructions	m	m			
Message Type	m	m			
Multi-part Body	m	m			
Non-receipt Notification Request Indication	m	c[1]			
Obsoleting Indication	m	m			
Originator Indication	m	m			
Originator Reference	m	m			
Originator PLAD	o	m	ACP127	m	
Other Recipient Indicator	m	m			
Pilot Forwarded	o	m	ACP127	m	
Primary and Copy Recipients Indication	m	m			
Receipt Notification Request Indication[2]	m	m			
Reply Request Indication	m	m			
Replying Message Indication	m	m			
Sensitivity Indication[3]	i	i			
Use of Address List	m	m			

NOTES

1 If any condition that could result in an NRN is supported, origination for NRN must also be supported. If an implementation can request a Non-receipt notification, it must be able to display it for the user if received.

2 If a Receipt Notification is received, the implementation must be able to display it for the user.

3 The originator's user interface shall not prompt for this EoS. If the protocol that supports this EoS is present upon reception then the EoS shall not be displayed to the user.

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Annex B

(normative)

AMENDMENTS AND CORRIGENDA

International Standards are subject to constant review and revision by the ISO/IEC Technical Committees concerned. The following amendments and corrigenda are approved by ISO/IEC JTC1 and are considered as normative references in this part of MIL-STD 2045-17502.

NOTE - Corresponding corrigenda to the equivalent CCITT Recommendations are contained in the joint *MHS Implementors' Guide*, Version 11 (ITU Special Rapporteur's Group on Message Handling Systems and ISO/IEC JTC1/SC18/WG4 SWG on Messaging).

ISO/IEC 10021-1/Cor.1:1991

ISO/IEC 10021-1/Cor.2:1991

ISO/IEC 10021-1/Cor.3:1992

ISO/IEC 10021-1/Cor.4:1992

ISO/IEC 10021-1/Cor.5:1992

ISO/IEC 10021-1/Cor.6:1994

ISO/IEC 10021-2/Cor.1:1991

ISO/IEC 10021-2/Cor.2:1991

ISO/IEC 10021-2/Cor.3:1992

ISO/IEC 10021-2/Cor.4:1992

ISO/IEC 10021-2/Cor.5:1993

ISO/IEC 10021-2/Cor.6:1994

ISO/IEC 10021-2/Cor.7:1994

ISO/IEC 10021-4/Cor.1:1991

ISO/IEC 10021-4/Cor.2:1991

ISO/IEC 10021-4/Cor.3:1992

ISO/IEC 10021-4/Cor.4:1992

ISO/IEC 10021-4/Cor.5:1992

ISO/IEC 10021-4/Cor.6:1993

ISO/IEC 10021-5/Cor.1:1991

ISO/IEC 10021-5/Cor.2:1991

ISO/IEC 10021-5/Cor.3:1992

ISO/IEC 10021-5/Cor.4:1992

ISO/IEC 10021-5/Cor.5:1992

ISO/IEC 10021-5/Cor.6:1993

ISO/IEC 10021-5/Cor.7:1994

ISO/IEC 10021-7/Cor.1:1991

ISO/IEC 10021-7/Cor.2:1991

ISO/IEC 10021-7/Cor.3:1992

ISO/IEC 10021-7/Cor.4:1992

ISO/IEC 10021-7/Cor.5:1992

ISO/IEC 10021-7/Cor.6:1993

ISO/IEC 10021-7/Cor.7:1994

ISO/IEC 10021-7/Cor.8:1994

ISO/IEC 10021-1/Amd.1:1994

ISO/IEC 10021-2/Amd.1:1993

ISO/IEC 10021-2/Amd.2:1994

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Annex C

(informative)

CONCLUDING MATERIAL

C.1 Deviations from the Base Standards

The Defense Messaging System (DMS) program must satisfy the need for specialized features of the E-Mail system to support military unique features. A unique X.400 content type is required to satisfy military message handling and security requirements. This Military Message (MM) content type, P772, is based upon extensions to the X.400 Interpersonal Message (IPMS) content type, P22, and is detailed in the Allied Communications Publication 123 (ACP 123). ACP 123 is currently in draft form.

This profile defines the minimum static conformance requirements for a system implementing MM in a standard environment, in which bandwidth considerations are not a major concern, such as might be available for normal strategic command and control messaging. Some of the unique military requirements that go beyond the capabilities of IPMS include: non-routine precedence indications, interworking without loss of information or service with the current AUTODIN messaging system, information for onward distribution determination, guaranteed timely delivery, and a specified level of survivability and security.

Interoperability with NATO will be achieved because the protocol definition of the MM content type has been closely harmonized with the NATO Ad Hoc Working Group on Military Message Handling Systems (MMHS) and the same P772 is defined in STANAG 4406.

Additionally, Military Messages for the U.S. will be protected using the Message Security Protocol (MSP) content type. MSP implementations for protection of unclassified but sensitive messages will be specified by Multi-level Information System Security Initiative (MISSI). Both mechanisms are developed by the National Security Agency (NSA). The requirements for these security mechanisms will be detailed in additional DSPs and are outside the scope of this DSP.

C.2 Subject Term (Keyword) Listing

Allied Communication Publication (ACP)
Standardized Profile (DSP)
Data Communications Protocol Standards (DCPS)
DCPS Technical Management Panel (DTMP)
DSPICS Proforma
International Standardized Profile (ISP)
Message Handling System (MHS)
Message Transfer Agent (MTA)
Message Transfer System (MTS)
Military Messaging (P772)
Open Systems Interconnection (OSI)
PICS Proforma

C.3 Preparing Activity: Defense Information Systems Agency (DISA) - DC

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C.4 Reviewing Activity:

Army	SC
Air Force	02, 13, 17, 29, 90
DLA	DH
DMA	MP
DIA	DI
DOT	OST
OASD	IQ, DO, MA, IR
ODISC4	AC
STRICOM	PT
NAVY	EC, CH, ND, TD, OM
USMC	MC, CG

Project: DCPS-0001, Subproject 02

C.5 Custodians:

DISA:	DC
Army:	SC
Air Force:	90
Navy:	OM
DIA:	DI
NSA:	NS
USMC:	MC
DLA:	DH
Other:	Joint Staff/Architecture & Integration
	USSPACECOM

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-STD-2045-17502-1A	941130

3. DOCUMENT TITLE Information Technology-DOD Standardized Profiles AMH2n(D) Message Handling System (MHS) Military Messaging (P772) Part 1: MM MHS Service Support

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
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DD Form 1426, OCT 89 previous editions are obsolete. 198-290

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